

The inverse of a matrix. $A \in M_n(\mathbb{R})$.

$$[A : I_n] \xrightarrow{\text{elementary operations}} [I_n : A^{-1}] \quad (! \text{ no column operations})$$

If $X = (v_1, \dots, v_m)$ we say that the dimension of $\langle X \rangle$ is given by the rank of the matrix
the basis is given by the non-zero rows of the matrix.

④ $A = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{pmatrix}$, $A^{-1} = ?$

! Check that $\exists A^{-1}$.

$$\begin{pmatrix} 1 & -4 \\ -2 & -4 \\ -2 & -4 \end{pmatrix}$$

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 2 & 1 & 0 & 0 \\ 2 & 1 & -2 & 0 & 1 & 0 \\ 2 & -2 & 1 & 0 & 0 & 1 \end{array} \right) \xrightarrow{\substack{R_2 \leftarrow R_2 - 2R_1 \\ R_3 \leftarrow R_3 - 2R_1}} \left(\begin{array}{ccc|ccc} 1 & 2 & 2 & 1 & 0 & 0 \\ 0 & -3 & -6 & -2 & 1 & 0 \\ 0 & -6 & -3 & -2 & 0 & 1 \end{array} \right)$$

$$\xrightarrow{\substack{R_2 = -\frac{1}{3}R_2 \\ R_3 = -\frac{1}{3}R_3}} \left(\begin{array}{ccc|ccc} 1 & 2 & 2 & 1 & 0 & 0 \\ 0 & 1 & 2 & \frac{2}{3} & -\frac{1}{3} & 0 \\ 0 & 2 & 1 & \frac{2}{3} & 0 & -\frac{1}{3} \end{array} \right) \xrightarrow{\substack{R_1 \leftarrow R_1 - 2R_2 \\ R_3 \leftarrow R_3 - 2R_2}}$$

$$= \left(\begin{array}{ccc|ccc} 1 & 0 & -2 & 1 - \frac{4}{3} & \frac{2}{3} & 0 \\ 0 & 1 & 2 & \frac{2}{3} & -\frac{1}{3} & 0 \\ 0 & 0 & -3 & -\frac{2}{3} & \frac{2}{3} & -\frac{1}{3} \end{array} \right) \xrightarrow{R_3 = -\frac{1}{3}R_3}$$

$$= \left(\begin{array}{ccc|ccc} 1 & 0 & -2 & -\frac{1}{3} & -\frac{2}{3} & 0 \\ 0 & 1 & 2 & \frac{2}{3} & -\frac{1}{3} & 0 \\ 0 & 0 & 1 & \frac{2}{9} & -\frac{2}{9} & \frac{1}{9} \end{array} \right) \xrightarrow{\substack{R_1 \leftarrow R_1 + 2R_3 \\ R_2 \leftarrow R_2 - 2R_3}}$$

$$= \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{9} & -\frac{2}{9} & \frac{2}{9} \\ 0 & 1 & 0 & \frac{2}{9} & -\frac{1}{9} & -\frac{2}{9} \\ 0 & 0 & 1 & \frac{2}{9} & -\frac{2}{9} & \frac{1}{9} \end{array} \right)$$

$\underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{I_3} \quad \underbrace{\begin{pmatrix} \frac{1}{9} & -\frac{2}{9} & \frac{2}{9} \\ \frac{2}{9} & -\frac{1}{9} & -\frac{2}{9} \\ \frac{2}{9} & -\frac{2}{9} & \frac{1}{9} \end{pmatrix}}_{A^{-1}}$

! Certain method for the next exam

①

⑤. $A = \begin{pmatrix} 1 & 4 & 2 \\ 3 & 3 & 1 \\ 3 & 0 & -1 \end{pmatrix}$, $A^{-1} = ?$

$$\begin{array}{r} 3 \times 2 = 6 \\ 6 \times 3 = 18 \\ 18 - 1 = 17 \\ 17 - 8 = 9 \\ 9 - 4 = 5 \\ 5 - 4 = 1 \end{array}$$

$$\left(\begin{array}{ccc|ccc} 1 & 4 & 2 & 1 & 0 & 0 \\ 3 & 3 & 1 & 0 & 1 & 0 \\ 3 & 0 & -1 & 0 & 0 & 1 \end{array} \right) \xrightarrow[R_3 \leftarrow R_3 - 3R_1]{R_2 \leftarrow R_2 - 2R_1} \left(\begin{array}{ccc|ccc} 1 & 4 & 2 & 1 & 0 & 0 \\ 1 & -5 & -3 & -2 & 1 & 0 \\ 0 & -12 & -7 & -3 & 0 & 1 \end{array} \right)$$

A invertible? $\det A = \begin{vmatrix} 1 & 4 & 2 \\ 3 & 3 & 1 \\ 3 & 0 & -1 \end{vmatrix} = \begin{vmatrix} 1 & 4 & 2 \\ 3 & 3 & 1 \\ 3 & 0 & -1 \end{vmatrix} = -3 + 24 - 18 = 3 \neq 0$

$$\left(\begin{array}{ccc|ccc} 1 & 4 & 2 & 1 & 0 & 0 \\ 3 & 3 & 1 & 0 & 1 & 0 \\ 3 & 0 & -1 & 0 & 0 & 1 \end{array} \right) \xrightarrow{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{ccc|ccc} 1 & 4 & 2 & 1 & 0 & 0 \\ 3 & 3 & 1 & 0 & 1 & 0 \\ 0 & -3 & -2 & 0 & -1 & 1 \end{array} \right)$$

$$\xrightarrow{R_3 \leftarrow R_3 + R_2} \left(\begin{array}{ccc|ccc} 1 & 4 & 2 & 1 & 0 & 0 \\ 3 & 3 & 1 & 0 & 1 & 0 \\ 0 & -3 & -2 & 0 & -1 & 1 \end{array} \right) \xrightarrow{R_3 \leftarrow \frac{1}{2}R_3} \left(\begin{array}{ccc|ccc} 1 & 4 & 2 & 1 & 0 & 0 \\ 3 & 3 & 1 & 0 & 1 & 0 \\ 0 & -1 & -1 & 0 & -\frac{1}{2} & \frac{1}{2} \end{array} \right)$$

$$\xrightarrow[R_3 \leftarrow R_3 - 2R_1]{R_2 \leftarrow R_2 - 3R_1} \left(\begin{array}{ccc|ccc} 1 & 4 & 2 & 1 & 0 & 0 \\ 0 & -5 & -7 & -3 & 1 & 0 \\ 0 & -7 & -5 & -2 & 0 & \frac{1}{2} \end{array} \right) \xrightarrow{L_3 \leftarrow L_3 - L_2} \left(\begin{array}{ccc|ccc} 1 & 4 & 2 & 1 & 0 & 0 \\ 0 & -5 & -7 & -3 & 1 & 0 \\ 0 & -2 & 2 & 1 & -1 & \frac{1}{2} \end{array} \right)$$

$$\xrightarrow{R_1 \leftarrow 2R_3 + R_1} \left(\begin{array}{ccc|ccc} 1 & 4 & 2 & 1 & 0 & 0 \\ 0 & -5 & -7 & -3 & 1 & 0 \\ 0 & -2 & 2 & 1 & -1 & \frac{1}{2} \end{array} \right) =$$

Path:

$$\begin{cases} R_2 = R_2 - 2R_1 \\ R_3 = R_3 - 3R_1 \\ R_3 = R_3 - 2R_2 \\ R_1 = R_1 + 2R_3 \\ R_3 = R_3 + 2R_2 \\ R_3 = -1R_3 \end{cases}$$

$$R_1 \leftarrow R_1 - R_3$$

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 0 & 0 & -1 & -\frac{1}{2} \\ 0 & -5 & -7 & -3 & 1 & 0 \\ 0 & -2 & 2 & 1 & -1 & \frac{1}{2} \end{array} \right)$$

$$R_1 \leftarrow R_1 + R_3$$

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & -2 & 0 \\ 0 & -5 & -7 & -3 & 1 & 0 \\ 0 & -2 & 2 & 1 & -1 & \frac{1}{2} \end{array} \right)$$

$$= ?$$

$$\begin{array}{l} 3 - 8 = -5 \\ 1 - 6 = -5 \\ 1 - 8 = -7 \\ -1 - 4 = -5 \\ 0 - \\ -5 - 7 = -12 \\ -7 + 5 = -2 \\ -7 - (-5) = -2 \\ -5 - (-7) = 2 \\ -2 + 3 = 1 \end{array}$$

6

$$B = (e_1, e_2, e_3, e_4)$$

$$X = (v_1, v_2, v_3)$$

$$v_1 = 3e_1 + 2e_2 - 5e_3 + 4e_4$$

$$v_2 = 3e_1 - e_2 + 3e_3 - 3e_4$$

$$v_3 = 3e_1 + 5e_2 - 13e_3 + 11e_4$$

$$[X]_B = ?$$

Det. its echelon form.

X is lin. indep.?

$$L_2 - L_1$$

$$L_3 - L_1$$

$$X = \begin{pmatrix} 3 & 2 & -5 & 4 \\ 3 & -1 & 3 & -3 \\ 3 & 5 & -13 & 11 \end{pmatrix} \xrightarrow{L_2 \leftarrow L_2 - L_1} \begin{pmatrix} 3 & 2 & -5 & 4 \\ 0 & -3 & 8 & -7 \\ 3 & 5 & -13 & 11 \end{pmatrix} =$$

$$\xrightarrow{P_3 \leftarrow P_3 - P_1} \begin{pmatrix} 3 & 2 & -5 & 4 \\ 0 & -3 & 8 & -7 \\ 0 & 3 & -8 & 7 \end{pmatrix} \xrightarrow{P_3 \leftarrow P_3 + P_2} \begin{pmatrix} 3 & 2 & -5 & 4 \\ 0 & -3 & 8 & -7 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\Rightarrow v_1, v_2, v_3 \text{ lin. dep.}$$

$$\Rightarrow \dim X = 2.$$

$$v_1, v_2 \text{ lin. indep.}$$

$$\Rightarrow \langle X \rangle = \langle v_1, v_2 \rangle. \text{ or}$$

7 $X = (v_1, v_2, v_3, v_4)$

$$v_1 = (1, 0, 4)$$

$$v_2 = (2, 1, 0)$$

$$v_3 = (1, 5, -36)$$

$$v_4 = (2, 10, -72)$$

$$\Rightarrow X = \begin{pmatrix} 1 & 0 & 4 \\ 2 & 1 & 0 \\ 1 & 5 & -36 \\ 2 & 10 & -72 \end{pmatrix} \xrightarrow{L_4 \leftarrow L_4 - 2L_3}$$

$$= \begin{pmatrix} 1 & 0 & 4 \\ 2 & 1 & 0 \\ 1 & 5 & -36 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{L_3 \leftarrow L_3 - L_1}$$

Det. a basis and dim for $\langle X \rangle$?

$$\Rightarrow v_1, v_2, v_3 \text{ lin. indep.}$$

$$\Rightarrow X = \langle (1, 0, 4), (2, 1, 0), (1, 5, -36) \rangle$$

$$\dim \langle X \rangle = 3.$$

$$\begin{cases} P_3 = P_3 - 9P_1 + 5P_2 \\ P_4 = P_4 - 18P_1 + 10P_2 \\ P_2 = P_2 - 2P_1 \end{cases}$$

$$\rightarrow \langle (1, 0, 4), (0, 1, -8) \rangle.$$

$$= \begin{pmatrix} 1 & 0 & 4 \\ 2 & 1 & 0 \\ 0 & -5 & -40 \\ 0 & 0 & 0 \end{pmatrix} \dots$$

3

$$\textcircled{8} \quad X = (v_1, v_2, v_3)$$

$$v_1 = (1, 0, 4, 3)$$

$$v_2 = (0, 2, 3, 1)$$

$$v_3 = (0, 4, 6, 2)$$

$$\Rightarrow \begin{pmatrix} 1 & 0 & 4 & 3 \\ 0 & 2 & 3 & 1 \\ 0 & 4 & 6 & 2 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - 2R_2} \begin{pmatrix} 1 & 0 & 4 & 3 \\ 0 & 2 & 3 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\Rightarrow \langle X \rangle = \langle (1, 0, 4, 3), (0, 2, 3, 1) \rangle$$

$$\Rightarrow \dim \langle X \rangle = 2.$$

basis?

$\dim \langle X \rangle$?

$$\textcircled{9} \quad S = \langle (1, 0, 4), (2, 1, 0), (1, 1, -4) \rangle.$$

$$T = \langle (-3, -2, 4), (5, 2, 4), (-2, 0, -8) \rangle$$

Det. a basis for $S, T, S+T$.

$\dim S, T, S+T, S \cap T$

$$S: \begin{pmatrix} 1 & 0 & 4 \\ 2 & 1 & 0 \\ 1 & 1 & -4 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 4 \\ 2 & 1 & 0 \\ 1 & 1 & -4 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & -8 \\ 1 & 1 & -4 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & -8 \\ 0 & 0 & -8 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 4 \\ 2 & 1 & 0 \\ 0 & 1 & -8 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 4 \\ 2 & 1 & 0 \\ 0 & 0 & -8 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & -8 \\ 0 & 0 & -8 \end{pmatrix}$$

$$\Rightarrow S = \langle (1, 0, 4), (0, 1, -8) \rangle. \quad \Rightarrow \dim \langle S \rangle = 2.$$

$$T: \begin{pmatrix} -3 & -2 & 4 \\ 5 & 2 & 4 \\ -2 & 0 & -8 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 + R_1 + R_3} \begin{pmatrix} -3 & -2 & 4 \\ 0 & 0 & 0 \\ -2 & 0 & -8 \end{pmatrix} \xrightarrow{R_2 = R_3} \begin{pmatrix} -3 & -2 & 4 \\ 0 & 0 & 0 \\ -2 & 0 & -8 \end{pmatrix}$$

$$= \begin{pmatrix} -3 & -2 & 4 \\ -2 & 0 & -8 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 + \frac{2}{3}R_1} \begin{pmatrix} -3 & -2 & 4 \\ 0 & \frac{4}{3} & \frac{16}{3} \\ 0 & 0 & 0 \end{pmatrix}$$

$$\Rightarrow T = \langle (-3, -2, 4), (0, \frac{4}{3}, \frac{16}{3}) \rangle$$

$$\Rightarrow \dim \langle T \rangle = 2$$

$$\begin{aligned} & -2 + \frac{4}{3} = -\frac{2}{3} \\ & 0 + \frac{16}{3} = \frac{16}{3} \\ & -8 - \frac{2}{3} = -\frac{24}{3} - \frac{2}{3} = -\frac{26}{3} \end{aligned} \quad \textcircled{4}$$

$$\begin{aligned}
 S+T &= \begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & -8 \\ -3 & -2 & 4 \\ 0 & 4 & 16 \end{pmatrix} \xrightarrow{-4R_2+R_4 \rightarrow R_4} \begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & -8 \\ -3 & -2 & 4 \\ 0 & 0 & 48 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 + 3R_2} \begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & -8 \\ 0 & 1 & -20 \\ 0 & 0 & 48 \end{pmatrix} \\
 &\quad +32+16 \\
 &\quad = 48 \\
 &\quad n+12 \\
 &= \begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & -8 \\ 0 & 0 & 48 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 0 & 4 \\ 0 & 0 & 48 \\ 0 & 1 & -8 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_3 \leftarrow 2R_2 + R_3} \begin{pmatrix} 1 & 0 & 4 \\ 0 & 0 & 48 \\ 0 & 1 & -8 \\ 0 & 0 & 0 \end{pmatrix}
 \end{aligned}$$

$$\Rightarrow \dim(S+T) = 3.$$

$$S+T = \langle (1, 0, 4), (0, 1, -8), (0, 0, 48) \rangle.$$

$$\dim S+T = \dim S$$

$$\dim S + \dim T = \dim(S+T) + \dim(S \cap T)$$

$$2 + 2 = 3 + \dim(S \cap T)$$

$$\Rightarrow \dim(S \cap T) = 1.$$

Summary

1. rank A?

$$\begin{aligned}
 & \begin{pmatrix} 0 & 2 & 3 \\ 2 & 4 & 3 \\ 1 & 1 & 1 \\ 2 & 2 & 4 \end{pmatrix} \xrightarrow{l_3 \leftrightarrow l_1} \begin{pmatrix} 1 & 1 & 1 \\ 2 & 4 & 3 \\ 0 & 2 & 3 \\ 2 & 2 & 4 \end{pmatrix} \xrightarrow{l_2 \leftarrow l_2 - 2l_1} \\
 & = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 2 & 3 \\ 2 & 2 & 4 \end{pmatrix} \xrightarrow{l_4 \leftarrow l_4 - 2l_1} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{pmatrix} \xrightarrow{l_3 = l_3 - l_2} \\
 & = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 2 \end{pmatrix} \xrightarrow{l_4 \leftarrow l_4 - l_3} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix} \rightarrow \text{one line of } 0 \\
 & \Rightarrow \text{rank} = 3.
 \end{aligned}$$

$$\begin{aligned}
 2. & \begin{pmatrix} 1 & -1 & 3 & 2 \\ -2 & 0 & 3 & -1 \\ -1 & 2 & 0 & -1 \end{pmatrix} \xrightarrow{l_2 \leftarrow l_2 + 2l_1} \begin{pmatrix} 1 & -1 & 3 & 2 \\ 0 & -2 & 9 & -3 \\ -1 & 2 & 0 & -1 \end{pmatrix} \xrightarrow{l_3 \leftarrow l_3 + l_1} \\
 & = \begin{pmatrix} 1 & -1 & 3 & 2 \\ 0 & -2 & 9 & -3 \\ 0 & 1 & 3 & 1 \end{pmatrix} \xrightarrow{l_3 \leftarrow 2l_3 + l_2} \begin{pmatrix} 1 & -1 & 3 & 2 \\ 0 & -2 & 9 & -3 \\ 0 & 0 & 15 & -1 \end{pmatrix} \xrightarrow{l_2 \leftarrow \frac{1}{2}(l_2 + l_3)} \\
 & = \begin{pmatrix} 1 & -1 & 3 & 2 \\ 0 & -1 & 12 & -2 \\ 0 & 0 & 15 & -1 \end{pmatrix} \xrightarrow{l_1 \leftarrow l_1 - l_3} \dots \begin{pmatrix} 1 & 0 & 0 & -3 \\ 0 & -1 & 0 & 4 \\ 0 & 0 & 3 & 5 \end{pmatrix} \xrightarrow{-5 \cdot \frac{1}{3}} \\
 & \downarrow \neq 0 \Rightarrow \text{rank} = 3.
 \end{aligned}$$

$$\begin{aligned}
 3. & \begin{pmatrix} \beta & 1 & 3 & 4 \\ 1 & \alpha & 3 & 3 \\ 2 & 3\alpha & 4 & 7 \end{pmatrix} \xrightarrow{l_2 \leftrightarrow l_1} \begin{pmatrix} 1 & \alpha & 3 & 3 \\ \beta & 1 & 3 & 4 \\ 2 & 3\alpha & 4 & 7 \end{pmatrix} \xrightarrow{l_3 \leftarrow l_3 - 3l_2} \begin{pmatrix} 1 & \alpha & 3 & 3 \\ \beta & 1 & 3 & 4 \\ -1 & 0 & -5 & -2 \end{pmatrix} \xrightarrow{l_2 \leftarrow l_2 + l_3} \\
 & = \begin{pmatrix} 1 & \alpha & 3 & 3 \\ \beta & 1 & 3 & 4 \\ 0 & \alpha & -2 & 2 \end{pmatrix} \xrightarrow{l_1 \leftarrow l_1 + l_3} \begin{pmatrix} \beta & 1 & 3 & 4 \\ 1 & \alpha & 3 & 3 \\ -1 & 0 & -5 & -2 \end{pmatrix} \xrightarrow{l_1 \leftarrow l_1 + l_2} \\
 & = \begin{pmatrix} \beta-1 & 1 & -2 & 2 \\ 0 & \alpha & -2 & 1 \\ -1 & 0 & -5 & -2 \end{pmatrix} \xrightarrow{l_1 \leftarrow l_1 + l_2} \begin{pmatrix} \beta-1 & 1-\alpha & 0 & 2 \\ 0 & \alpha & -2 & 1 \\ -1 & 0 & -5 & -2 \end{pmatrix} \xrightarrow{-\frac{1}{5}l_3 + l_1} \\
 & = \begin{pmatrix} \beta-1 & 1-\alpha & 0 & 2 \\ 0 & \alpha & -2 & 1 \\ 0 & 0 & -5 & -2 \end{pmatrix} \xrightarrow{-2+2} \dots
 \end{aligned}$$

$$4. \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 2 & | & 1 & 0 & 0 \\ 2 & 1 & -2 & | & 0 & 1 & 0 \\ 2 & -2 & 1 & | & 0 & 0 & 1 \end{pmatrix} \begin{array}{l} l_2 \leftarrow l_2 - 2l_1 \\ l_3 \leftarrow l_3 - 2l_1 \end{array}$$

$$= \begin{pmatrix} 1 & 2 & 2 & | & 1 & 0 & 0 \\ 0 & -3 & -6 & | & -2 & 1 & 0 \\ 0 & -6 & -3 & | & -2 & 0 & 1 \end{pmatrix} \begin{array}{l} l_3 \leftarrow l_3 - 2l_2 \\ -2 - 4 \end{array}$$

$$= \begin{pmatrix} 1 & 2 & 2 & | & 1 & 0 & 0 \\ 0 & -3 & -6 & | & -2 & 1 & 0 \\ 0 & 0 & 9 & | & 2 & -2 & 1 \end{pmatrix} \begin{array}{l} l_1 \leftarrow 3l_1 + 2l_2 \\ -6 + 6 \\ -3 + 12 \\ -2 + 4 \end{array}$$

$$= \begin{pmatrix} 3 & 0 & 6 & | & -1 & 2 & 0 \\ 0 & -3 & -6 & | & -2 & 1 & 0 \\ 0 & 0 & 9 & | & 2 & -2 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & -2 & | & -\frac{1}{3} & \frac{2}{3} & 0 \\ 0 & 1 & 2 & | & \frac{2}{3} & -\frac{1}{3} & 0 \\ 0 & 0 & 1 & | & \frac{2}{9} & -\frac{2}{9} & \frac{1}{9} \end{pmatrix} \begin{array}{l} l_2 \leftarrow l_2 - 2l_3 \\ l_1 \leftarrow l_1 + 2l_3 \end{array}$$

$$= \begin{pmatrix} 1 & 0 & 0 & | & \frac{1}{3} & \frac{2}{3} & \frac{2}{9} \\ 0 & 1 & 0 & | & \frac{2}{3} & -\frac{1}{3} & \frac{2}{9} \\ 0 & 0 & 1 & | & \frac{2}{9} & -\frac{2}{9} & \frac{1}{9} \end{pmatrix}$$

↓
id. matrix

↓
the inverse.

5. -

6) $K, B = (e_1, e_2, e_3, e_4)$ basis

$$\lambda = (v_1, v_2, v_3) \text{ test.} \rightarrow X = \begin{pmatrix} 3 & 2 & -5 & 4 \\ 3 & -1 & 3 & -3 \\ 3 & 5 & -13 & 11 \end{pmatrix} \xrightarrow{l_2 - l_1 \rightarrow l_2}$$

$N_1 = \dots$

$$= \begin{pmatrix} 3 & 2 & -5 & 4 \\ 0 & -3 & 8 & -7 \\ 3 & 5 & -13 & 11 \end{pmatrix} \xrightarrow{l_3 - l_1 \rightarrow l_3} \begin{pmatrix} 3 & 2 & -5 & 4 \\ 0 & -3 & 8 & -7 \\ 0 & 3 & -8 & 7 \end{pmatrix} \xrightarrow{l_3 + l_2 \rightarrow l_3} \begin{matrix} -13+5 \end{matrix}$$

$$= \begin{pmatrix} 3 & 2 & -5 & 4 \\ 0 & -3 & 8 & -7 \\ 0 & 0 & 0 & 0 \end{pmatrix} \Rightarrow v_1, v_2, v_3 \text{ lin. dep.}$$

7) $X = (v_1, \dots)$

$\dim \langle X \rangle$

a basis of $\langle X \rangle$.

We want zeroes.

$$\begin{pmatrix} 1 & 0 & 4 \\ 2 & 1 & 0 \\ 1 & 5 & -36 \\ 2 & 10 & -72 \end{pmatrix} \xrightarrow{l_4 \leftarrow l_4 - 2l_3} \begin{pmatrix} 1 & 0 & 4 \\ 2 & 1 & 0 \\ 1 & 5 & -36 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{l_2 \leftarrow l_2 - 2l_1} \begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & -8 \\ 1 & 5 & -36 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow{l_3 \leftarrow l_3 - l_1} \begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & -8 \\ 0 & 5 & -40 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{l_3 \leftarrow -5l_2 + l_3} \begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & -8 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$\Rightarrow \text{rank} X = 2$

$\Rightarrow \dim \langle X \rangle = 2 \Rightarrow \langle X \rangle = \langle v_1, v_2 \rangle$

8) -

9

$$\dim S \rightarrow b_1$$

$$\dim T \rightarrow b_1$$

$$\dim(S+T) \rightarrow b_?$$

$$\dim(S \cap T)$$

$$S = \langle (1, 0, 4), (2, 1, 0), (1, 1, -4) \rangle$$

$$T = \langle (-3, -2, 4), (5, 2, 4), (-2, 0, -8) \rangle$$

$$S: \begin{pmatrix} 1 & 0 & 4 \\ 2 & 1 & 0 \\ 1 & 1 & -4 \end{pmatrix} \xrightarrow{l_3 \leftarrow l_1 + l_3} \begin{pmatrix} 2 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 1 & -4 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 4 \\ 2 & 1 & 0 \\ 2 & 1 & 0 \end{pmatrix} \xrightarrow{l_3 \leftarrow l_3 - l_2}$$

$$= \begin{pmatrix} 1 & 0 & 4 \\ 2 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \dim S = 2 \Rightarrow S = \langle (1, 0, 4), (2, 1, 0) \rangle.$$

$$T: \begin{pmatrix} -3 & -2 & 4 \\ 5 & 2 & 4 \\ -2 & 0 & -8 \end{pmatrix} \xrightarrow{l_2 \leftarrow l_1 + l_2} \begin{pmatrix} -3 & -2 & 4 \\ 2 & 0 & 8 \\ -2 & 0 & -8 \end{pmatrix} \xrightarrow{l_3 \leftarrow l_2 + l_3}$$

$$= \begin{pmatrix} -3 & -2 & 4 \\ 2 & 0 & 8 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow \dim T = 2 \Rightarrow T = \langle (-3, -2, 4), (5, 2, 4) \rangle.$$

$$\dim(S+T) = \dim \langle S \cup T \rangle$$

$$\dim(S+T) = \dim S + \dim T - \dim(S \cap T)$$

$$S+T: \begin{pmatrix} 1 & 0 & 4 \\ 2 & 1 & 0 \\ 1 & 1 & -4 \\ -3 & -2 & 4 \\ 5 & 2 & 4 \\ -2 & 0 & -8 \end{pmatrix} \Rightarrow \dim(S+T) = 3$$

$$\Rightarrow \langle S+T \rangle = \dots$$

$$\begin{pmatrix} \beta & 1 & 3 & 4 \\ 1 & \alpha & 3 & 3 \\ 2 & 3\alpha & 4 & 7 \end{pmatrix} \xrightarrow{L_3 \leftarrow L_2 + L_3} \begin{pmatrix} \beta & 1 & 3 & 4 \\ 1 & \alpha & 3 & 3 \\ 0 & \alpha & -2 & 1 \end{pmatrix}$$

$$\xrightarrow{L_3 \leftarrow L_3 - L_2} \begin{pmatrix} \beta & 1 & 3 & 4 \\ 1 & \alpha & 3 & 3 \end{pmatrix}$$

$$\begin{pmatrix} \beta & 1 & 3 & 4 \\ 1 & \alpha & 3 & 3 \\ 2 & 3\alpha & 4 & 7 \end{pmatrix} \xrightarrow{L_3 \leftarrow L_3 - L_2} \begin{pmatrix} \beta & 1 & 3 & 4 \\ 1 & \alpha & 3 & 3 \\ 1 & 2\alpha & 1 & 7 \end{pmatrix} \quad \beta + 1, \alpha + 1, 6$$

$$\xrightarrow{L_3 \leftarrow L_3 - 2L_2} \begin{pmatrix} \beta & 1 & 3 & 4 \\ 1 & \alpha & 3 & 3 \\ -1 & 0 & -5 & 7 \end{pmatrix} \xrightarrow{L_2 \leftarrow L_2 + L_3} \begin{pmatrix} \beta & 1 & 3 & 4 \\ 0 & \alpha & -2 & 10 \\ -1 & 0 & -5 & 7 \end{pmatrix}$$

$$\xrightarrow{L_1 \leftarrow L_1 + L_3} \begin{pmatrix} \beta - 1 & 1 & -2 & 4 \\ 0 & \alpha & -2 & 10 \\ -1 & 0 & -5 & 7 \end{pmatrix}$$

$$\xrightarrow{L_1 \leftarrow L_1 + L_2} \begin{pmatrix} \beta - 1 & \alpha + 1 & 0 & 14 \\ 0 & \alpha & -2 & 10 \\ -1 & 0 & -5 & 7 \end{pmatrix} =$$

$$\begin{pmatrix} \beta & 1 & 3 & 4 \\ 1 & \alpha & 3 & 3 \\ 2 & 3\alpha & 4 & 7 \end{pmatrix} \xrightarrow{L_2 \leftarrow L_1 + L_2} \begin{pmatrix} \beta & 1 & 3 & 4 \\ \beta + 1 & \alpha + 1 & 6 & 7 \\ 2 & 3\alpha & 4 & 7 \end{pmatrix} = \begin{pmatrix} \beta + 4 & 1 & 3 & 4 \\ \alpha + 4 & \alpha & 3 & 3 \\ 3\alpha + 6 & 3\alpha & 4 & 7 \end{pmatrix}$$

123
 $12\alpha 1$

$$\xrightarrow{L_2 \leftarrow L_2 - L_3} \begin{pmatrix} \beta & 1 & 3 & 4 \\ -1 - 2\alpha & -1 & -4 & -4 \\ 2 & 3\alpha & 4 & 7 \end{pmatrix} =$$

$$\xrightarrow{L_3 \leftarrow L_3 - L_2} \begin{pmatrix} \beta & 1 & 3 & 4 \\ 1 & \alpha & 3 & 3 \\ 1 & 2\alpha & 1 & 4 \end{pmatrix}$$

$$= \begin{pmatrix} \beta + 4 & 1 & 3 & 4 \\ \alpha - \beta & \alpha - 1 & 0 & -1 \\ 3\alpha + 6 & 3\alpha & 4 & 7 \end{pmatrix}$$